

Seizure Disorders

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Seizure Disorders

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Objectives

To know what a seizure disorder is and its various types; the mechanisms of seizures; and the treatment and preventive measures.

Outline

Definition of terms · Epidemiology · Classification · Aetiology · Pathophysiology · Clinical features · Diagnosis · Treatment · Seizure mimics · Status epilepticus · Febrile seizures · Prevention · References

Definition of terms

- **Seizure** – a **discrete clinical event** (behavioural, emotional, motor or sensory) that reflects a **temporary physiologic dysfunction of the brain**, characterized by **excessive and hyper-synchronous discharge of cortical neurons**.
- **Seizure disorder** – a general term including any of several disorders: **epilepsy, febrile seizures**, and possibly single seizures and symptomatic seizures secondary to metabolic, infectious or other aetiologies.
- **Febrile seizures** – seizures occurring between the ages of **6 and 60 months (peak 12-18 months)** with a temperature of **38°C (100.4°F) or higher**, that are **not** the result of CNS infection or metabolic imbalance, and occur in the **absence of a prior afebrile seizure**.
- **Provoked seizure** – occurs secondary to an acute problem (e.g. high fever, infection). With elimination of the stimulus, the seizure stops.
- **Unprovoked seizure** – no obvious stimulus; a brain disorder producing recurrent spontaneous paroxysmal discharges.
- **Epilepsy** – a **chronic disorder** characterized by seizures that **recur unpredictably in the absence of a consistent provoking factor**.
- **Epileptic syndrome** – a disorder manifesting as one or more specific seizure types, with a **specific age of onset** and a **specific prognosis**.
- **Genetic epilepsy** (formerly idiopathic) – seizures without an identifiable cause, in a patient with **normal neurologic findings and normal intelligence**.
- **Structural epilepsy** (formerly symptomatic) – an epilepsy syndrome caused by an **underlying structural brain disorder**.

Convulsion (involuntary motor seizure): tonic (sustained contraction), clonic (broken/jerky), tonic-clonic. Other forms: **sensory** (abnormal sensations), **autonomic** (salivation, palpitation), **psychic** (abnormal fear, often with altered consciousness).

Epidemiology

- Seizures are the **most common neurological problem** affecting children worldwide
- Occur in **10% of children; fewer than 1/3 are caused by epilepsy**
- Cumulative **lifetime incidence of epilepsy is 3%** (over half begin in childhood)
- **Annual prevalence of epilepsy is lower, 0.5-0.8%** (many children outgrow it)
- Annual incidence: **40-70/100,000** in industrialized countries; **100-190/100,000** in resource-poor countries
- Rates are greatest in **early childhood**; slight male excess (**M:F = 1.5:1**); more common in **lower socio-economic groups**

Classification

By seizure type – 2017 ILAE operational classification

Divides seizures into **four categories** by the presumed mode of onset:

1. **Focal-onset**
2. **Generalized-onset**
3. **Unknown onset**
4. **Unclassified**

Focal-onset	Generalized-onset	Unknown-onset
Motor: tonic, clonic, atonic, myoclonic, hyperkinetic, epileptic spasms, automatisms	Motor: tonic-clonic, tonic, clonic, atonic, myoclonic, myoclonic-atonic, myoclonic-tonic-clonic, epileptic spasms	Motor: tonic-clonic, epileptic spasms
Non-motor: behaviour arrest, sensory, cognitive, emotional, autonomic Awareness: aware / impaired awareness Focal to bilateral tonic-clonic (formerly secondary generalized)	Non-motor (absence): typical, atypical, myoclonic, eyelid myoclonia	Non-motor: behaviour arrest

- **Generalized seizures** – clinical and EEG changes indicate **synchronous involvement of both hemispheres**
- **Focal seizures** – activation of a system of neurons **limited to part of one cerebral hemisphere**
- **Unknown onset** – insufficient information to determine focal or generalized
- **Unclassified** – unusual features, onset cannot be determined despite adequate workup

By syndrome

Based on: **age of onset** · **aetiology (if known)** · **seizure type** · **genetics** · **natural history (drug response and remission)** · **EEG pattern**.

Aetiology

A seizure is a symptom, not a specific diagnosis.

Congenital

- **CNS malformation** – neurocutaneous syndromes (neurofibromatosis, tuberous sclerosis, Sturge-Weber), cerebral dysgenesis (porencephaly)
- **Genetic syndromes**, intrauterine infections

Acquired

- **Trauma** – birth trauma (causes both seizures and CP), asphyxia
- **Infections** – meningitis, encephalitis, cerebral malaria
- **Febrile seizure**

- **Metabolic** – deranged Na (high or low), low Ca, Mg, glucose
- **Toxic** – lead poisoning, drugs (intoxication, addiction or withdrawal)
- **CNS neoplasm**
- **Vascular** – AVM, CVA

Pathophysiology

Physiology of nerve impulse propagation:

- **Excitation** – glutamate and aspartate
- **Inhibition** – GABA (gamma-aminobutyric acid)

Both excitatory and inhibitory mechanisms act to **allow appropriate firing and prevent inappropriate excitation** of the cell.

Seizures occur due to:

- **Excitation** of the neuron – by excessive, frequent or sustained activation of **glutamate** receptors, increasing brain excitability
- **Inhibition failure** – disruption in **GABA** production and its receptors
- Or a **combination** of both

Four sequential mechanistic processes: (1) underlying aetiology; (2) epileptogenesis; (3) epileptic state of increased excitability; (4) seizure-related neuronal injury.

Clinical features

Tonic and clonic phases of a generalized seizure

Focal-onset (formerly partial) seizures

Have a **focus** (a localized part of the brain).

- **Simple partial** – mainly from frontal, parietal and occipital lobes
- **Complex partial** – mainly from the temporal lobe (also frontal, parietal)
- **EEG**: spikes and sharp waves; unilateral, bilateral or multifocal

Focal onset, awareness intact (formerly simple partial):

- Mostly **motor** activity – tonic or clonic (face, neck, extremities)
- **Versive seizures** are common – head turning and conjugate eye movement (indicating a lesion in the **opposite** frontal lobe)
- $\pm$ **aura** (warning); duration **10-20 sec**; **no automatism, no postictal phase, consciousness intact**

Focal to bilateral (psychomotor/temporal lobe, formerly complex partial):

- **Aura** indicating focal onset; **impaired consciousness**
- **Automatic behaviour** in 50-70%; duration **1-2 minutes**; a **postictal phase**; may generalize to a tonic-clonic convulsion

Generalized tonic-clonic seizure (GTCS)

The tonic-clonic sequence and its EEG

- The **commonest** clinical manifestation

- **Loss of tone and consciousness**; generalized tonic phase followed by **rhythmic clonic** contraction alternating with relaxation of all muscle groups
- $\pm$ autonomic effects – incontinence, apnoea, cyanosis
- **EEG (ictal)**: burst of spikes and polyspikes at **10-15 Hz** in the tonic phase; **diffuse high-voltage slow waves** in the clonic phase

EEG of a generalized tonic-clonic seizure

Benign childhood epilepsy with centrotemporal spikes (Rolandic epilepsy)

- Usually in **primary school children**; **remits in adolescence**
- Responds to drugs; **DOC is carbamazepine**
- Almost always occurs **at night** (partial to generalized)
- **EEG**: central and temporal spikes on a normal background

Absence seizure (simple or typical)

EEG of absence seizure – 3 Hz spike and wave

- **Sudden cessation of activity** with a blank facial expression
- Lasts **< 30 sec**; $\pm$ countless seizures/day
- Age of onset **5 years**; more in girls
- **No loss of tone, no postictal phase** – the child resumes pre-seizure activity; $\pm$ automatism
- **Atypical** – motor activity and loss of tone
- **EEG: 3/sec spike and generalized wave discharge**

Infantile spasm (IS)

EEG of hypsarrhythmia

- Age of onset **4-8 months**
- Types: flexion, extension (least common), **mixed (commonest)**
- Lasts **< 1 sec**; several episodes/day
- Cause: **80-90% symptomatic** (pre/peri/postnatal), 10-20% cryptogenic
- **80-90% risk of mental retardation**
- **EEG: hypsarrhythmia** – chaotic pattern of high-voltage, bilaterally asynchronous slow-wave activity
- **West syndrome** – the triad of **infantile spasms, mental retardation and hypsarrhythmia**

Diagnosis

The diagnosis of epilepsy is ultimately clinical. It is based on a detailed description of events by the patient or an eyewitness.

Determine: is it epilepsy or are the seizures related to an **acute encephalopathy**; if epilepsy, **categorize** the seizure/syndrome; and find any **identifiable treatable aetiology**.

Investigations

- **FBC/ESR** (on clinical indication), **RBS, S/E/U/Cr** including **Ca and Mg**
- **Anticonvulsant blood levels** assay
- Blood and urine **amino acids**, urine **organic acids**
- **EEG** including awake and sleep tracings; special studies – 24-hour ambulatory, video-EEG, all-night recording

- **ECG** if a cardiac disorder is suspected
- **Skull X-rays** if abnormal head shape/size
- **Neuroimaging** (CT/MRI, PET, SPECT) – evidence of raised ICP, focal neurologic signs
- **LP** if infection is suspected and there are **no contraindications**

Differential diagnosis / seizure mimics

Conditions sharing a sudden onset of abnormal consciousness, behaviour, posture, tone, sensation or autonomic function:

- **Breath-holding spells** (cyanotic and pallid)
- **Gastro-oesophageal reflux** (in small infants)
- **Benign paroxysmal vertigo**
- **Syncope** (simple and cough) – resembles the atonic type
- **Night terrors, tics**
- **Narcolepsy, cataplexy, sleep apnoea**
- **Hypoglycaemia**
- **Parasomnias** – night terrors, sleep walking, sleep talking, nocturnal enuresis
- **Prolonged QT syndrome** – sudden loss of consciousness during exercise or emotional stress
- **Pseudoseizures / psychogenic non-epileptic seizures (PNES)** – cause is psychological

Treatment

Stepwise approach:

- **Step 1** – ABC of resuscitation (airway, ventilation, cardiac function, TPR, BP, RBS, quick history)
- **Step 2** – abort the seizure (using the status epilepticus guideline)
- **Step 3** – detailed evaluation (history and examination to elicit the cause)
- **Step 4** – treat any identifiable cause
- **Step 5** – for epilepsy or an epileptic syndrome, place on an appropriate anti-seizure medication (ASM) and continue follow-up in the neurology clinic

Ketogenic diet – high-fat, low-carbohydrate, low-protein diet for severe seizure disorders.

Surgery – for symptomatic localization-related epilepsy refractory to ASM; resection of lesions (tumour, AVM) or lobectomy (e.g. anterior temporal lobectomy).

Antiepileptic drugs

Drug	Type of epilepsy	Mechanism	Dose	Side effects
Phenobarbitone	GTC, partial	Opens Cl channel via GABA receptor	3-5 mg/kg/day PO	Hyperkinesia, drowsiness
Phenytoin	GTC, partial, status	Blocks Na ⁺ -dependent channels & depolarization-dependent Ca ²⁺ uptake	5-10 mg/kg/day PO	Gum hypertrophy, acne, ataxia, nystagmus

Drug	Type of epilepsy	Mechanism	Dose	Side effects
Carbamazepine	GTC, partial	Same as phenytoin	10-20 mg/kg/day PO	Rashes, ataxia, low Hb, neutropenia, hepatotoxicity
Ethosuximide	Typical absence	Blocks Ca ²⁺ channel of thalamocortical circuitry	20-30 mg/kg/day PO	Vomiting, rashes
Na valproate (broad spectrum)	GTC, absence, myoclonic	Blocks voltage-dependent Na ⁺ channels & Ca ²⁺ -dependent K ⁺ conductance	20-30 mg/kg/day PO	Nausea, drowsiness, tremor, epigastric pain
ACTH	Infantile spasm	Suppresses CRH secretion by feedback	20 U/day	High glucose, high BP, cushingoid facies
Benzodiazepines (clonazepam, diazepam, lorazepam)	GTC, myoclonic, absence, status	Bind a GABA site enhancing Cl channel opening	Clonazepam 0.1-0.2 mg/kg/day PO; diazepam 0.25-0.3 mg/kg IV or 0.5 mg/kg rectal; lorazepam 0.1 mg/kg IV	CNS depression

Status epilepticus

- **Old definition** – a seizure prolonged for **> 30 minutes**, or repeated seizures with no recovery of consciousness in between
- **Current definition** – a seizure lasting **longer than 5 minutes**, or more than one seizure within a 5-minute period without returning to normal consciousness

Generalized tonic-clonic is the most alarming because of the risk of hypoxia.

Aetiology: febrile seizure, non-compliance with drugs, stress including intercurrent infection, brain pathology (tumour, trauma, infection, infarction, drugs).

Management of status epilepticus

- **ABC of resuscitation;** blood sugar estimation; IV line, blood sample for investigation
- **First line – benzodiazepines** (S/E: respiratory depression, hypotension): IV diazepam **0.1-0.3 mg/kg** (max 3 doses), IV lorazepam **0.05-0.1 mg/kg** slowly; for difficult IV access – rectal diazepam **0.2-0.5 mg/kg**, or buccal/nasal midazolam **0.5 mg/kg**
- **Second line** (if the seizure persists): phenytoin **15-20 mg/kg** (max 30; fosphenytoin preferred – S/E arrhythmia, ECG monitoring); phenobarbital **15-20 mg/kg**; Na valproate IV **20-40 mg/kg** over 10 min (max 1200 mg); levetiracetam **40 mg/kg** over 5 min (max 3 g)
- **Third line** (ICU care): induction of anaesthesia with thiopental, midazolam, pentobarbital or propofol; intubate and ventilate; continuous EEG and cardiopulmonary monitoring; treat any identifiable cause

Febrile seizures

- Occur in **2-5% of neurologically healthy infants and children**; commoner in **males**
- **Hereditary** – tends to run in families (possibly autosomal dominant)
- Common causes of the fever: **malaria**, upper respiratory tract infection, otitis media, sore throat/pharyngitis

Classification of febrile seizures

Simple febrile seizure	Complex febrile seizure
Brief, < 15 minutes	> 15 minutes
Usually single episode in 24 hours	Usually multiple in 24 hours
Generalized	Focal

Diagnosis and management

Before diagnosing a febrile seizure, rule out intracranial infection.

Management: ABCD of resuscitation; control the temperature with antipyretics; abort the seizure with **diazepam** (IV slowly to avoid respiratory arrest, 0.1-0.3 mg/kg; IV or rectal) or **intramuscular paraldehyde** (0.1 ml/kg).

Investigate for the cause: FBC, blood film for malaria parasites, and **lumbar puncture for CSF analysis** – to rule out meningitis, **especially in children < 18 months** who do not show the usual signs.

Treat the underlying cause. **Counsel parents** on fever management at home and reassure them that the disorder is **benign** but can recur.

Risk factors for recurrence of febrile seizures

- **Major:** age < 1 year; duration of fever < 24 hr; fever 38-39°C
- **Minor:** family history of febrile seizures; family history of epilepsy; complex febrile seizure; daycare; male gender; lower serum sodium at presentation

Risk of subsequent epilepsy after a febrile seizure

Risk factor	Risk of subsequent epilepsy
Simple febrile seizure	1%
Recurrent febrile seizures	4%
Complex febrile seizures (>15 min or recurrent in 24 hr)	6%
Fever < 1 hr before the seizure	11%
Family history of epilepsy	18%
Complex febrile seizures (focal)	29%
Neurodevelopmental abnormalities	33%

Prevention

1. **General health promotion** – health education, good health-seeking behaviour, optimal infant feeding/nutrition, immunization, hygiene, female education

2. **Specific protection** – avoid triggers for those with a predisposition or family history
3. **Prompt diagnosis and treatment** – appropriate evidence-based management
4. **Limitation of disability** – compliance with ASM, treatment of identified treatable conditions
5. **Rehabilitation** – physiotherapy, special school, mobility devices, psychotherapy

References

Nelson Textbook of Paediatrics (Kliegman, Stanton, St. Geme, Schor, Behrman); Textbook of Paediatrics and Child Health in a Tropical Region (Azubuike and Nkanginieme).